

Technology Stack

Date	15 July 2026
Team ID	XXXXXX
Project Name	AI Debt Settlement System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3, Tailwind CSS	Provides a modern, responsive, and component-based user interface with reusable components and faster development.
2	Backend / Server-Side	FastAPI (Python)	High-performance Python framework for building RESTful APIs with automatic documentation, validation, and asynchronous support.
3	Programming Language	Python, JavaScript	Python is used for backend logic, AI integration, and database operations, while JavaScript powers the React frontend.
4	Database	SQLite	Lightweight relational database suitable for development, testing, and small-to-medium applications without requiring a separate database server.

5	ORM (Object Relational Mapper)	SQLAlchemy	Simplifies database operations using Python objects, reducing SQL complexity and improving maintainability.
6	Authentication & Security	JWT (JSON Web Token), Passlib (bcrypt)	Provides secure user authentication, password hashing, and authorization for protected APIs.
7	AI / Machine Learning API	Google Gemini AI API	Generates AI-powered financial analysis, debt stress assessment, settlement recommendations, and negotiation strategies.
8	HTTP Client	Axios	Enables seamless communication between the React frontend and FastAPI backend using REST APIs.
9	PDF Generation	jsPDF	Generates downloadable AI settlement reports in PDF format directly from the frontend.
10	API Testing	Swagger UI (FastAPI Docs)	Automatically generated API documentation for testing and verifying backend endpoints.
11	Development Environment	Visual Studio Code	Lightweight IDE with excellent support for Python, React, FastAPI, debugging, and Git integration.
12	Version Control	Git & GitHub	Tracks source code changes, enables collaboration, and hosts the project repository for deployment and evaluation.

